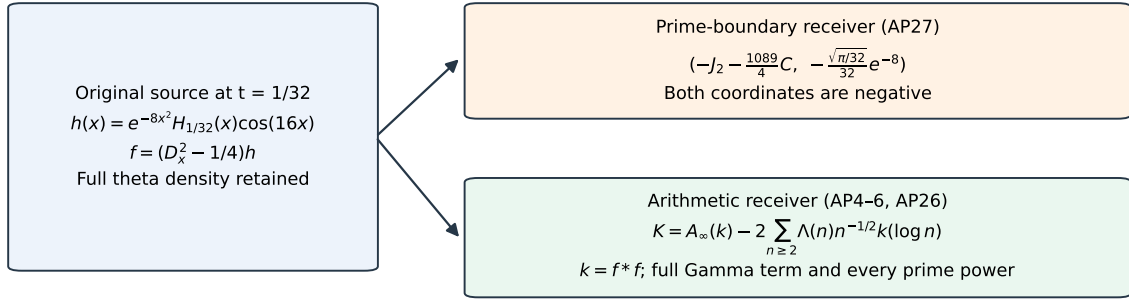


An evaluated arithmetic pairing for the original endpoint test

Split-Zero research programme

23 September 2026



The two arrows are different proved maps of the same source. No support label is removed.

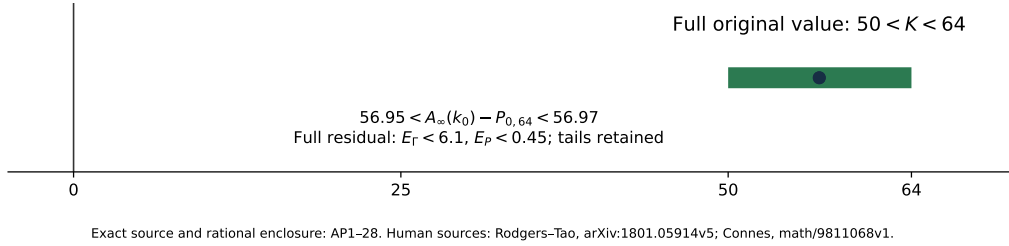


Figure 1: The same original source has the two distinct, fully computed receivers AP27 and AP4-6. The green interval is the certified enclosure AP1, including the entire residual AP26. The dark point marks the narrow comparison interval AP25. The diagram displays exact formulas and proved rational bounds, rather than sampled zeta zeros.

1 The actual endpoint test has a strictly positive arithmetic pairing

This calculation evaluates the test constructed by the endpoint-resonance calculation. It uses the original meromorphic Riemann zeta function, its prime powers, its pole, every trivial zero at each finite contour cutoff, and the full Gamma boundary. At the original time $t = 1/32$, the result is the certified rational enclosure

$$50 < K_{1/32} < 64. \quad (\text{AP1})$$

Here K_t is defined below for this one explicit test. The two negative source moments of the same test therefore do not give a negative Weil pairing. The calculation determines this test's sign; it does not assert positivity for every test.

1.1 Original source, coordinates, and zeta factors

Throughout, x, u are the original real coordinates and $t = 1/32$. Retain

$$\begin{aligned}\Phi(u) &= \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \\ H(x) &= \int_0^{\infty} e^{u^2/32} \Phi(u) \cos(xu) du, \\ h(x) &= e^{-8x^2} H(x) \cos(16x), \quad f(x) = h''(x) - \frac{1}{4}h(x), \quad k = f * f, \\ F(s) &= \int_{\mathbb{R}} f(x) e^{-(s-1/2)x} dx.\end{aligned}\tag{AP2}$$

The density and heat convention are those of Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, [version 5](#), original TeX equations `phidef`, `htdef`, `hoz`, and `sas`. Only those source formulas are used here; the present sign calculation uses no assumption about the location of zeta zeros.

For clarity, the test's exact return to the original function is displayed with every factor. Set $w = s - 1/2$, $\sigma_0 = itw$, $\sigma_1 = 1 + itw$. Then

$$\begin{aligned}F(s) &= \frac{s(s-1)\sqrt{\pi t}}{16} \left\{ e^{t(w-i/(2t))^2} \sigma_1(\sigma_1-1) \pi^{-\sigma_1/2} \Gamma(\sigma_1/2) \zeta(\sigma_1) \right. \\ &\quad \left. + e^{t(w+i/(2t))^2} \sigma_0(\sigma_0-1) \pi^{-\sigma_0/2} \Gamma(\sigma_0/2) \zeta(\sigma_0) \right\}.\end{aligned}\tag{AP3}$$

Every product at an exceptional argument is the analytic continuation of the displayed full product. In particular the products at arguments zero and one both equal one. The pole of ζ at one and the pole of $\Gamma(s/2)$ at zero are included in this statement. To prove AP3, Gaussian integration gives

$$\int_{\mathbb{R}} e^{-x^2/(4t)} H_t(x) e^{-wx} dx = 2\sqrt{\pi t} e^{tw^2} H_0(2tw).$$

The two exponential terms of $\cos(x/(2t))$, followed by $16H_0(z) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ with $s = 1/2 + iz/2$, give AP3. Absolute interchange follows from the outer Gaussian and the super-exponential original density. Two integrations by parts give the retained factor $s(s-1)$. Consequently $F(0) = F(1) = 0$, exactly.

1.2 Complete arithmetic receiver

Write $j = \zeta'/\zeta$, and let ρ run through the nontrivial zeros of the original function with multiplicity m_ρ . The exact quantity evaluated is

$$K = \sum_{\rho} m_\rho F(\rho)^2 = A_\infty(k) - P(k), \quad P(k) = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} k(\log n),\tag{AP4}$$

where $\Lambda(p^a) = \log p$ for every prime power and $\Lambda(n) = 0$ otherwise. The full archimedean term is

$$A_\infty(k) = -(\gamma_E + \log \pi)k(0) + \int_0^\infty \frac{e^{-x}k(0) - e^{-x/4}k(x/2)}{1 - e^{-x}} dx.\tag{AP5}$$

Evenness and reality give $\overline{F(1-\bar{s})} = F(s)$. Thus AP4 has the actual reflected Weil product; away from the critical line it has not been replaced by an absolute square.

We recall the finite-cutoff derivation to retain the original divisor. Put $A = F^2$, $b_N = -2N - 1$, $U_N = \sum_{m=1}^N A(-2m)$, and $\kappa(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \psi_\Gamma(s/2)$. For upward vertical integrals

$I_c = (2\pi i)^{-1} \int_{\Re s=c}$, the argument principle and the full original functional equation give

$$\begin{aligned} V_N &= K + U_N - A(1) = I_\sigma(Aj) - I_{b_N}(Aj), \quad \sigma > 1, \\ j(s) + j(1-s) &= -\kappa(s) - \kappa(1-s), \\ V_N &= -P(k) + \mathcal{G}_N, \\ \mathcal{G}_N &= I_{b_N}\{A(s)[\kappa(s) + \kappa(1-s)]\} = A_\infty(k) + A(0) + U_N. \end{aligned} \tag{AP6}$$

The rational derivatives of $s(s-1)$ in the original multiplier cancel with their reflected partners in the displayed reflection identity; the pole at one remains explicitly in V_N . On the right edge $j(s) = -\sum \Lambda(n)n^{-s}$; Mellin inversion gives the two prime legs. Moving the Gamma edge to $1/2$ crosses exactly the residues at $0, -2, \dots, -2N$, giving $A(0) + U_N$. The digamma integral on $1/2$ gives AP5 with its combined numerator. This proves AP4 because both endpoints vanish. No limit or finite value is assigned to U_N individually.

For the classical contour and measure conventions, see Alain Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, [Appendix II, \(1\)–\(11\)](#), [Theorems 1 and 6](#). Its compact-support theorem is extended here by the actual Gaussian bounds below. They give arbitrary inverse-power decay of F on fixed vertical strips by integration by parts. The original zero count $O(T \log(T+2))$ gives absolute convergence of AP4, and contours at heights avoiding zero ordinates have vanishing horizontal integrals. All strips used in AP6 have finite width.

1.3 An exact decomposition with the entire residual retained

Define the positive original moments

$$J_j = \int_0^\infty u^j e^{u^2/32} \Phi(u) du, \quad C = J_0, \quad R(x) = H(x) - C. \tag{AP7}$$

Every summand of Φ is positive for $u \geq 0$. The inequalities $e^{4u} \geq 1 + 4u + 8u^2$, $\pi > 3$, $\pi^2 < 10$ imply

$$J_j \leq 20j! \sum_{n \geq 1} \frac{n^4 e^{-3n^2}}{(12n^2 - 9)^{j+1}} < \frac{j!}{3^{j+1}}. \tag{AP8}$$

Indeed $12n^2 - 9 \geq 3n^2$, and $\sum n^2 e^{-3n^2} < 1/20$. One fully rational check of the last inequality uses $e^3 > 2007/100$, $q = 1/8000$, and

$$\sum n^2 e^{-3n^2} < \frac{100}{2007} + \frac{1}{20} \left(\frac{1+q}{(1-q)^3} - 1 \right) < \frac{1}{20}.$$

Differentiation under the integral is therefore valid. In particular

$$|R(x)| \leq J_2 x^2/2, \quad |R'(x)| \leq J_2 |x|, \quad |R''(x)| \leq J_2, \quad |R'''(x)| \leq J_3. \tag{AP9}$$

Put $f = f_0 + e_f$, where the equality, rather than a replacement of f , defines the residual:

$$f_0(x) = C e^{-8x^2} [(256x^2 - 1089/4) \cos(16x) + 512x \sin(16x)]. \tag{AP10}$$

Direct differentiation of AP2 gives

$$\begin{aligned} e_f(x) &= e^{-8x^2} \{ [(256x^2 - 1089/4)R - 32xR' + R''] \cos(16x) \\ &\quad + [512xR - 32R'] \sin(16x) \}. \end{aligned} \tag{AP11}$$

No term of this residual is dropped from the pairing. With $r \geq 0$, define

$$\begin{aligned} A(r) &= 1089/4 + 512r + 256r^2, \\ E(r) &= 1 + 32r + (1345/8)r^2 + 256r^3 + 128r^4, \\ B(r) &= 4868 + 13060r + 12288r^2 + 4096r^3, \\ E_1(r) &= 48 + (3457/4)r + 3970r^2 + 7298r^3 + 6144r^4 + 2048r^5. \end{aligned} \tag{AP12}$$

AP9–11 and their derivatives prove

$$\begin{aligned} |f_0(x)| &\leq Ce^{-8x^2}A(|x|), \quad |e_f(x)| \leq J_2e^{-8x^2}E(|x|), \\ |f'_0(x)| &\leq Ce^{-8x^2}B(|x|), \quad |e'_f(x)| \leq e^{-8x^2}[J_2E_1(|x|) + J_3]. \end{aligned} \tag{AP13}$$

For an explicit derivative check, the cosine coefficient in $e^{8x^2}e'_f$ is $(13060x - 4096x^3)R + (768x^2 - 3265/4)R' - 48xR'' + R'''$; its sine coefficient is $(4868 - 12288x^2)R + 1536xR' - 48R''$. This proves every coefficient in AP12 without omitting the derivative of the original source.

1.4 Exact Gaussian convolution and uniform residual bounds

Write $k_0 = f_0 * f_0$ and $\Delta = k - k_0$. The exact convolution is

$$\begin{aligned} k_0(x) &= \frac{C^2\sqrt{\pi}}{128}e^{-4x^2}\{(65536x^4 - 1622528x^2 + 1250369)\cos(16x) \\ &\quad + 2048x(256x^2 - 1121)\sin(16x) \\ &\quad + e^{-16}(65536x^4 - 49664x^2 + 3137)\}. \end{aligned} \tag{AP14}$$

To verify it, put $h_0(x) = Ce^{-8x^2}\cos(16x)$. In the convolution set $y = x/2 + z$ and use the full product identity $\cos(16y)\cos(16(x-y)) = [\cos(16x) + \cos(32z)]/2$. The Gaussian integrals then give exactly

$$(h_0 * h_0)(x) = \frac{C^2\sqrt{\pi}}{8}e^{-4x^2}[\cos(16x) + e^{-16}].$$

Since $f_0 = (D^2 - 1/4)h_0$, differentiation under the convolution gives $k_0 = (D^2 - 1/4)^2(h_0 * h_0)$. Expanding these two derivatives yields all coefficients in AP14, including the opposite-frequency contribution e^{-16} .

For any polynomial P with nonnegative coefficients put

$$\mathcal{I}(P) = \int_{\mathbb{R}} e^{-16x^2}P(|x|)dx, \quad \mathcal{I}(r^j) = 16^{-(j+1)/2}\Gamma((j+1)/2). \tag{AP15}$$

The substitution $v = 16x^2$ on the two half-lines proves this identity with all constants. Define the explicit upper bounds

$$\begin{aligned} n_0 &= C\sqrt{\mathcal{I}(A^2)}, \quad n_e = J_2\sqrt{\mathcal{I}(E^2)}, \\ n_1 &= C\sqrt{\mathcal{I}(B^2)}, \quad n_{e1} = \sqrt{J_2^2\mathcal{I}(E_1^2) + 2J_2J_3\mathcal{I}(E_1) + J_3^2\mathcal{I}(1)}, \\ d_0 &= 2n_0n_e + n_e^2, \quad d_2 = 2n_1n_{e1} + n_{e1}^2. \end{aligned} \tag{AP16}$$

Since $\Delta = 2f_0 * e_f + e_f * e_f$, Cauchy–Schwarz and $k'' = f' * f'$ give

$$|\Delta(x)| \leq d_0, \quad |\Delta''(x)| \leq d_2, \quad |\Delta(x) - \Delta(0)| \leq \min(2d_0, d_2x^2/2). \tag{AP17}$$

The last bound uses evenness. For the arithmetic sum there is also the stronger pointwise envelope

$$|\Delta(x)| \leq e^{-4x^2}\mathcal{I}(2CJ_2A(r + |x|/2)E(r + |x|/2) + J_2^2E(r + |x|/2)^2) =: \mathcal{D}(x). \tag{AP18}$$

Indeed, after $y = x/2 + z$, both $|y|$ and $|x - y|$ are at most $|z| + |x|/2$. All coefficients in AP12 are positive. This proves AP18 directly for the complete residual convolution.

1.5 Every infinite tail and the Gamma endpoint

The actual source bound AP8 gives, by differentiating AP2,

$$|f(x)| \leq (95 + 175|x| + 86x^2)e^{-8x^2} \leq (183 + 174x^2)e^{-8x^2}.$$

The same bound holds for f_0 , since $C < 1/3$. Convolving the last polynomial and retaining the Gaussian yields

$$|k(x)|, |k_0(x)| \leq Q(x)e^{-4x^2}, \quad Q(x) = 16000 + 6900x^2 + 850x^4. \quad (\text{AP19})$$

For verification the polynomial before the integer upper bounds is

$$\frac{\sqrt{\pi}}{4}[9105363/256 + (247167/16)x^2 + (7569/4)x^4];$$

$\sqrt{\pi}/4 < 4/9$ proves AP19 coefficient by coefficient.

For $L \geq 4$ put

$$T_p(L, c) = \sum_{j=0}^p \binom{p}{j} L^{p-j} j! / c^{j+1}.$$

Monotonicity of the function $v \mapsto (\log v)Q(\log v)v^{-1/2}e^{-4(\log v)^2}$ on $v \geq 64$ gives the prime-power tail bound

$$2 \sum_{n > 64} \frac{\Lambda(n)}{\sqrt{n}} |k(\log n)| \leq 2 \int_4^\infty u Q(u) e^{u/2-4u^2} du < 10^{-21}. \quad (\text{AP20})$$

To check monotonicity, the logarithmic derivative in u is $1/u + Q'(u)/Q(u) - 1/2 - 8u$, with $Q'/Q \leq 4/u$. Use $\Lambda(n) \leq \log n$, then the decreasing sum-to-integral bound and $\log 64 > 4$. Expanding at $u = 4 + v$ and using $e^{-4v^2} \leq 1$ bounds the integral by

$$2e^{-62}[16000T_1(4, 63/2) + 6900T_3(4, 63/2) + 850T_5(4, 63/2)] < 100000e^{-62} < 10^{-21};$$

the bracket with its factor two is 1882788609056000/20841167403. The last inequality uses $e^3 > 20$. No prime power below the cutoff is omitted.

For either convolution, the exact Gamma tail is

$$\int_8^\infty \frac{e^{-x}k(0) - e^{-x/4}k(x/2)}{1 - e^{-x}} dx = -k(0)\log(1 - e^{-8}) - \mathcal{R}_8, \quad |\mathcal{R}_8| < 10^{-23}. \quad (\text{AP21})$$

The first term is integrated exactly. AP19 bounds the other term by

$$\frac{e^{-66}}{1 - e^{-8}}[16000T_0(8, 65/4) + 1725T_2(8, 65/4) + (425/8)T_4(8, 65/4)].$$

The bracket is 1006957513344/46411625, its double is below 100000, and $e^{66} > 20^{22} > 10^{28}$, proving AP21.

At zero the combined Gamma integrand tends to $-3k(0)/4$. For k_0 , its integral over $[0, \epsilon]$, $\epsilon = 1/65536$, is retained within the symmetric bound

$$B_{\text{origin}} = \frac{3}{2}k_0(0)\epsilon + n_1^2\epsilon^2/8. \quad (\text{AP22})$$

In fact $1 - e^{-x} \geq x/2$, $|e^{-x} - e^{-x/4}| \leq 3x/4$ on $[0, 1]$, and $|k_0(x/2) - k_0(0)| \leq n_1^2x^2/8$. The integrand is bounded by $3k_0(0)/2 + n_1^2x/4$, which integrates to AP22.

For the full residual choose $b = 1/16$. The same combined numerator, split over $[0, b]$, $[b, 1]$, and $[1, \infty)$, proves

$$|A_\infty(k) - A_\infty(k_0)| \leq E_\Gamma := \left[\gamma_E + \log \pi + \frac{3}{2} + 4\log(1/b) + \frac{e^{-1} + 4e^{-1/4}}{1 - e^{-1}} \right] d_0 + d_2b^2/8. \quad (\text{AP23})$$

On $[0, 1]$ the part with $\Delta(0)$ costs at most $3d_0/2$. The remaining part costs $d_2b^2/8$ below b and $4d_0\log(1/b)$ above it. For $x \geq 1$, bound both numerator terms by d_0 times their original exponentials and use $1 - e^{-x} \geq 1 - e^{-1}$. This proves AP23.

1.6 Finite interval certificate and conclusion

The source moments are integrated with the original summands $1 \leq n \leq 16$, $0 \leq u \leq 3$, with radius 10^{-300} added for the complete omitted source. Here is the tail proof. For $u = 2 + v$, $v \geq 0$, use $e^8 > 400$ to obtain

$$u^2/32 + 9u - \pi n^2 e^{4u} \leq 145/8 - 1200n^2 - (4800n^2 - 73/8)v - (9600n^2 - 1/32)v^2 \leq -1180n^2 - v.$$

For $j \leq 4$, $(2 + v)^j \leq (2 + v)^4$, whose integral against e^{-v} is 168. The omitted part with $u \geq 2$ is therefore at most $3360 \sum n^4 e^{-1180n^2}$. Since $n^2 \geq 1 + 3(n - 1)$, the geometric fourth-moment sum at $q = e^{-3540} < 1/2$ is less than 300, giving $1008000e^{-1180} < 10^{-383}$.

For the index tail apply AP8 only to $n \geq 17$. Write $n = 17 + a$; then $n^2 \geq 289 + 35a$. For $q = e^{-105} < 1/2$,

$$\sum_{n \geq 17} n^2 e^{-3n^2} \leq e^{-867} \left[\frac{289}{1 - q} + \frac{34q}{(1 - q)^2} + \frac{q(1 + q)}{(1 - q)^3} \right] < 652e^{-867}.$$

For $j = 0, 2, 3, 4$, $20j!/3^{j+1} \leq 20/3$, so that tail is below $10000e^{-867} < 10^{-369}$. The last comparisons follow from $e^3 > 20$ and $2^{10} > 10^3$. The union of these two omitted regions contains the required omitted domain, proving the stated radius. All original summands are positive, so there is no uncontrolled cancellation in the moment-tail estimate.

The executable `certify_pairing.py` uses 100-bit ball arithmetic and `acb.integral`, with 24 subintervals for each source moment and 64 for AP5 on $[\epsilon, 8]$. Its integrands contain only entire functions and field operations; a possible division by zero returns an infinite ball. Thus no unsupported branch is declared analytic. All resulting enclosures are finite. The integration contract and error bounds are documented in the original [FLINT/Arb integration documentation](#), implemented by Fredrik Johansson and the FLINT contributors. The bounds use the resulting interval radii, not the requested tolerance alone.

The retained output gives

$$\begin{aligned} 0.06216254 < C < 0.06216255, \quad 0 < J_2 < 0.000719, \quad 0 < J_3 < 0.000120, \\ d_0 < 0.303, \quad d_2 < 175, \quad E_\Gamma < 6.1, \\ E_P := 2 \sum_{2 \leq n \leq 64} \frac{\Lambda(n)}{\sqrt{n}} \mathcal{D}(\log n) < 0.45. \end{aligned} \tag{AP24}$$

Every entry follows from finite sums of AP15 and ball integration of the explicitly displayed functions. The certificate retains the more precise intervals and checks each stated rational comparison.

Define $P_{0,64}$ using AP4 with k_0 and $n \leq 64$. AP14, the integrated Gamma interval, AP21, and AP22 give

$$56.95 < A_\infty(k_0) - P_{0,64} < 56.97. \tag{AP25}$$

Finally the exact equality

$$K = A_\infty(k_0) - P_{0,64} + [A_\infty(k) - A_\infty(k_0)] - [P_{64}(k) - P_{0,64}] - P_{>64}(k) \tag{AP26}$$

retains the entire source correction. AP20 and AP23–25 imply

$$56.95 - 6.1 - 0.45 - 10^{-21} < K < 56.97 + 6.1 + 0.45 + 10^{-21},$$

which proves AP1. No zero table, spectral truncation, presumed critical-line property, or deformed Euler product was used.

1.7 The source information carried by this sign

The original endpoint-null filter is invertible on Schwartz space:

$$h(x) = - \int_{\mathbb{R}} e^{-|x-y|/2} f(y) dy.$$

The derivative jump gives $(D^2 - 1/4)(-e^{-|x|/2}) = \delta_0$; polynomially weighted convolution bounds prove continuity on Schwartz space. The homogeneous solutions $e^{x/2}, e^{-x/2}$ have no nonzero Schwartz combination. Thus the filter retains the original source. Its two prime-boundary coordinates, in the receiver [FR15–25](#), are

$$\beta[(t_p - 1) \otimes f] = \ell_p \otimes \left(-J_2 - \frac{1089}{4}C, -\frac{\sqrt{\pi/32}}{32}e^{-8} \right). \quad (\text{AP27})$$

The first entry follows by differentiating AP2 at zero; the second follows by integrating f and evaluating AP3 at $s = 1/2$. Both entries are negative and nonzero, while AP1 is positive. These statements concern exactly the same source and the proved maps.

Every original support label is retained: for a bounded distributive lattice L , use

$$G_L(V) = \{(v, 1_L) : v \in V\} \cup \{(0, \lambda) : \lambda \neq 1_L\}, \quad e = (0, 1_L), \quad \tau = (0, 0_L). \quad (\text{AP28})$$

The linear source and receiver maps send $(v, 1_L)$ to $(Av, 1_L)$ and fix each lower zero label. The quadratic receiver sends a top-supported input to $(K(f), 1_L)$ and keeps the label of each zero-amplitude input. This is a specified labelled quadratic map, not a semiring homomorphism. No Boolean quotient is imposed. AP1 evaluates the amplitude of this map on the actual top-supported test.